

WEEK-7 SKILL — Student Records CRUD API

POST:

REST client interface showing a POST request to `http://localhost:3000/students`. The request body is a JSON object:

```
1 {
2   "name": Harshika
3   "age": 18,
4   "course": "CSIT",
5   "email": mallipeddiharshika@klh.edu.in
6 }
```

The response is a 201 Created status with a JSON body:

```
1 {
2   "name": Harshika
3   "age": 18,
4   "course": "CSIT",
5   "email": mallipeddiharshika@klh.edu.in
6   "_id": "6ab0d6e94bd72bb2efd336e4",
7   "createdAt": "2026-09-21T07:04:09.524Z",
8   "updatedAt": "2026-09-21T07:04:09.524Z",
9   "__v": 0
10 }
```

After POST in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: Harshika
    age: 18,
    course: 'CSIT',
    email: mallipeddiharshika@klh.edu.in
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  }
]
```

GET all data:

REST client interface showing a GET request to `http://localhost:3000/students`. The response is a 200 OK status with a JSON array:

```
1 [
2   {
3     "_id": "6ab0d6e94bd72bb2efd336e4",
4     "name": Harshika
5     "age": 18,
6     "course": "CSIT",
7     "email": mallipeddiharshika@klh.edu.in
8     "createdAt": "2026-09-21T07:04:09.524Z",
9     "updatedAt": "2026-09-21T07:04:09.524Z",
10    "__v": 0
11  },
12  {
13    "_id": "6ab0d7b74bd72bb2efd336e5",
14    "name": "K Venkat Ram Reddy",
15    "age": 23,
16    "course": "MS in BA",
17    "email": "kvrreddy@gmail.com",
18    "createdAt": "2026-09-21T07:07:35.746Z",
19    "updatedAt": "2026-09-21T07:07:35.746Z",
20    "__v": 0
21  }
22 ]
```

GET all data in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: Harshika
    age: 18,
    course: 'CSIT',
    email: mallipeddiharshika@klh.edu.in
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  },
  {
    _id: ObjectId('6ab0d7b74bd72bb2efd336e5'),
    name: 'K Venkat Ram Reddy',
    age: 23,
    course: 'MS in BA',
    email: 'kvrreddy@gmail.com',
    createdAt: ISODate('2026-09-21T07:07:35.746Z'),
    updatedAt: ISODate('2026-09-21T07:07:35.746Z'),
    __v: 0
  }
]
```

GET one with specific id:

The screenshot shows a REST client interface with a GET request to `http://localhost:3000/students/6ab0d6e94bd72bb2efd336e4`. The response is a 200 OK status with a response time of 9 ms and a body size of 433 B. The response body is displayed in JSON format, showing the details of the student Harshika.

```
{
  "_id": "6ab0d6e94bd72bb2efd336e4",
  "name": "Harshika",
  "age": 18,
  "course": "CSIT",
  "email": "mallipeddiharshika@klh.edu.in",
  "createdAt": "2026-09-21T07:04:09.524Z",
  "updatedAt": "2026-09-21T07:04:09.524Z",
  "__v": 0
}
```

PATCH → changed age from 23 to 25:

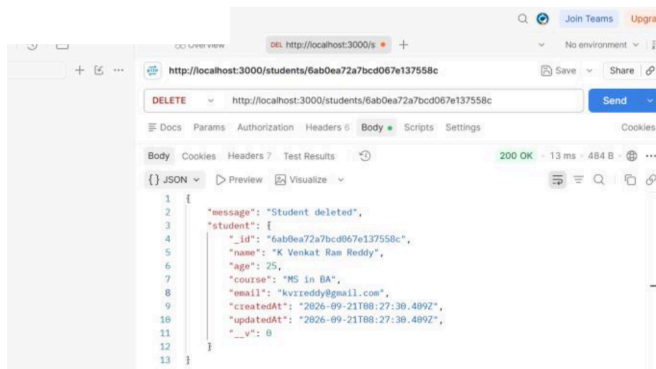
The screenshot shows a REST client interface with a PATCH request to `http://localhost:3000/students/6ab0d7b74bd72bb2efd336e5`. The request body is a JSON object with the field `"age": 25`. The response is a 200 OK status with a response time of 594 ms and a body size of 444 B. The response body is displayed in JSON format, showing the updated details of the student K Venkat Ram Reddy.

```
{
  "_id": "6ab0d7b74bd72bb2efd336e5",
  "name": "K Venkat Ram Reddy",
  "age": 25,
  "course": "MS in BA",
  "email": "kvrreddy@gmail.com",
  "createdAt": "2026-09-21T07:07:35.746Z",
  "updatedAt": "2026-09-21T08:24:56.224Z",
  "__v": 0
}
```

After modification of age output in mongosh:

```
{
  _id: ObjectId('6ab0ea72a7bcd067e137558c'),
  name: 'K Venkat Ram Reddy',
  age: 25,
  course: 'MS in BA',
  email: 'kvrreddy@gmail.com',
  createdAt: ISODate('2026-09-21T08:27:30.409Z'),
  updatedAt: ISODate('2026-09-21T08:27:30.409Z'),
  __v: 0
}
```

DELETE:



Observation of output in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: Harshika
    age: 18,
    course: 'CSIT',
    email: mallipediharshika@klh.edu.in
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  }
]
student_records>
```

Folder Structure:

Week_7_StudentRecordsAPI

- ├── controllers
 - └── studentController.js
- ├── models
 - └── Student.js
- ├── routes
 - └── studentRoutes.js
- └── server.js

└─ package.json

Code:

package.json

```
{
  "name": "week_7_studentrecordsapi",
  "version": "1.0.0",
  "type": "module",
  "description": "",
  "main": "server.js",
  "scripts": {
    "start": "node server.js",
    "dev": "nodemon server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.2.1",
    "mongoose": "^9.10.1"
  },
  "devDependencies": {
    "nodemon": "^3.1.14"
  }
}
```

models/Student.js

```
import mongoose from 'mongoose';

const studentSchema = new mongoose.Schema({
  name: {
    type: String,
    required: true,
    trim: true
  },
  age: {
    type: Number,
    required: true,
    min: 16,
    max: 100
  },
  course: {
    type: String,
    required: true,
    trim: true
  },
  email: {
    type: String,
```

```

        required: true,
        unique: true,
        lowercase: true
    }
}, {
    timestamps: true
});

const Student = mongoose.model('Student', studentSchema);

export default Student;

```

controllers/studentController.js

```

import Student from '../models/Student.js';

export const createStudent = async (req, res) => {
    try {
        const student = await Student.create(req.body);
        res.status(201).json(student);
    } catch (error) {
        res.status(400).json({ error: error.message });
    }
};

export const getAllStudents = async (req, res) => {
    try {
        const students = await Student.find();
        res.status(200).json(students);
    } catch (error) {
        res.status(500).json({ error: error.message });
    }
};

export const getStudentById = async (req, res) => {
    try {
        const student = await Student.findById(req.params.id);

        if (!student)
            return res.status(404).json({ message: 'Student not found' });

        res.status(200).json(student);
    } catch (error) {
        res.status(400).json({ error: error.message });
    }
};

export const updateStudent = async (req, res) => {
    try {

```

```

const student = await Student.findByIdAndUpdate(
  req.params.id,
  req.body,
  { new: true, runValidators: true }
);

if (!student)
  return res.status(404).json({ message: 'Student not found' });

res.status(200).json(student);
} catch (error) {
  res.status(400).json({ error: error.message });
}
};

export const deleteStudent = async (req, res) => {
  try {
    const student = await Student.findByIdAndDelete(req.params.id);

    if (!student)
      return res.status(404).json({ message: 'Student not found' });

    res.status(200).json({
      message: 'Student deleted',
      student
    });
  } catch (error) {
    res.status(400).json({ error: error.message });
  }
};

```

routes/studentRoutes.js

```

import express from 'express';

import {
  createStudent,
  getAllStudents,
  getStudentById,
  updateStudent,
  deleteStudent
} from '../controllers/studentController.js';

const router = express.Router();

router.post('/', createStudent);
router.get('/', getAllStudents);
router.get('/:id', getStudentById);

```

```
router.patch('/:id', updateStudent);
router.delete('/:id', deleteStudent);
```

```
export default router;
```

server.js

```
import express from 'express';
import mongoose from 'mongoose';
import studentRoutes from './routes/studentRoutes.js';

const app = express();
const PORT = 3000;

app.use(express.json());

app.get('/', (req, res) => {
  res.json({
    message: 'Student Records API is running'
  });
});

app.use('/students', studentRoutes);

mongoose.connect('mongodb://127.0.0.1:27017/student_records')
  .then(() => {
    console.log('MongoDB connected');

    app.listen(PORT, () => {
      console.log(`Server running on http://localhost:${PORT}`);
    });
  })
  .catch((error) => {
    console.error('MongoDB connection failed:', error.message);
  });
```